

SEAF-KG: Structurally-Enhanced Agentic Fusion for Insurance Knowledge Graphs

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1 Pipeline Overview

SEAF-KG is a 4-zone pipeline that automatically builds a knowledge graph with an induced domain ontology from heterogeneous insurance data:

1. **Zone 1 — Ingestion:** PDFs $\rightarrow$ text chunks; CSVs $\rightarrow$ structured records with schema detection.
2. **Zone 2 — Extraction:** Dual-path approach — LLM extraction for PDFs (Qwen 2.5 72B), deterministic structured mapper for CSVs. Followed by entity resolution and data-driven property collapse.
3. **Zone 3 — SV-LOI Ontology Induction:** Novel 7-stage method that fuses LLM semantic typing with graph-structural verification to induce domain ontology classes.
4. **Zone 4 — Storage:** Neo4j graph database with INSTANCE_OF, SUBCLASS_OF, and ASSOCIATED_WITH edges.

2 Dataset

We evaluate on the Emory Spring 2026 dataset provided by Assurant, spanning three insurance lines:

Table 1: Data sources

Source	Type	Rows	Domain
GEICO renters policies details	CSV	100	Home/Renters
GEICO renters claims	CSV	100	Home/Renters
GEICO renters survey (3 files)	CSV	300	Home/Renters
T-Mobile survey sample	CSV	100	Mobile/Device
T-Mobile claim sample	CSV	100	Mobile/Device
T-Mobile chat survey	CSV	100	Mobile/Device
Auto Service form	PDF	—	Auto

Reference ontology: **Riskine** (open-source insurance ontology, 26 classes, 46 edges).

3 Evaluation

3.1 Zone 2: Extraction Quality

Table 2: Extraction quality metrics

Metric	Score	Description
Triple Precision	91.1%	Factual correctness of extracted triples
Fact Recall	73.2%	Fraction of source facts captured in KG
Source Grounding	80.0%	Triples traceable to source document

Triple Precision. An LLM judge evaluates a sample of 100 extracted triples as Correct, Incorrect, or Uncertain. We scored 82 correct, 10 uncertain, 8 incorrect. Example:

- *Correct:* SERVICE CONTRACT --COVERS--> Heated Back Glass
- *Incorrect:* Claim Management --CAN_BE_CANCELED_AT--> mytmoclaim.com

Fact Recall. For 30 sampled source chunks, we extract 8 key facts per chunk and check whether matching triples exist in the KG using BERTScore cosine similarity (> 0.65 threshold). Result: 175 of 239 facts found (73.2%).

Source Grounding. For 50 sampled triples, we verify the source document supports the claim. 39 fully supported, 2 partially supported, 9 had no source found.

3.2 Zone 3: Ontology Quality

All metrics compare against the Riskine reference ontology (26 classes, 46 edges).

Table 3: Ontology quality metrics

Metric	Precision	Recall	F1
Riskine Class Alignment	0.750	0.385	0.508
BERTScore	0.726	0.564	0.635
Graph F1 (SGC, 2-hop)	0.745	0.373	0.497
Entity Assignment (full 26-class)	0.208	0.192	0.200
Entity Assignment (present 8-class)	0.208	0.625	0.312
Wu-Palmer Similarity	0.747 (13 pairs)		

BERTScore F1 (0.635). Measures semantic similarity between our induced class names and Riskine reference classes using BERT embeddings. For each induced class c_i , we compute cosine similarity against every reference class r_j and find the best match. Precision averages the best match per induced class; recall averages the best match per reference class:

$$P = \frac{1}{|C|} \sum_i \max_j \cos(c_i, r_j), \quad R = \frac{1}{|R|} \sum_j \max_i \cos(c_i, r_j)$$

This captures whether we discovered the right concepts even with different names. Example matches:

Induced	Riskine	Cosine	Verdict
Organization	Organization	1.00	Match
Policy	Product	0.90	Match
Person	Person	1.00	Match
Coverage	Coverage	1.00	Match
Property	Property	1.00	Match
Service	BusinessProcess	0.86	Partial
Risk	Address	0.80	Partial

Graph F1 (0.497). Structural comparison using Soft Graph Correspondence (SGC) with 2-hop neighborhoods. For each class node, we compare its local graph topology against the reference. Our precision is high (74.5%) — the edges we created are correct. Recall is lower (37.3%) because Riskine has 46 edges and we induced 8 SUBCLASS_OF + 38 ASSOCIATED_WITH, many of which don’t overlap with Riskine’s specific edge patterns.

Wu-Palmer Similarity (0.747). Measures taxonomy distance in a shared class hierarchy:

$$\text{WP}(c_1, c_2) = \frac{2 \cdot \text{depth}(\text{LCA}(c_1, c_2))}{\text{depth}(c_1) + \text{depth}(c_2)}$$

where LCA is the lowest common ancestor. Range: 0.0 (unrelated) to 1.0 (same node). Our score of 0.747 over 13 matched pairs indicates reasonable structural alignment with the reference hierarchy.

Entity Assignment F1 (0.200 full / 0.312 present). For each induced class, we embed all member entities, compute the class centroid, and find the nearest Riskine class by cosine distance.

- **Full 26-class** (F1=0.200): measured against all 26 Riskine classes, including those absent from our data (BankAccount, DrivingLicense, etc.). Structurally low recall.
- **Present 8-class** (F1=0.312): only against classes with evidence in the data. Better reflects actual alignment.

Example assignments: Coverage → Coverage (cos=0.67, score=1.0), Risk → Risk (cos=0.45, score=0.5), Person → Identification (cos=0.45, partial).

3.3 Why Entity Assignment F1 Is Low

The full 26-class Entity Assignment F1 (0.200) deserves explanation. Riskine defines 26 classes, but our dataset only contains evidence for 7–8 of them. The remaining 17–18 classes (e.g., BankAccount, CreditCard, DrivingLicense, Education, Animal, Vehicle) simply do not appear in GEICO renters, T-Mobile device, or auto service data. This means recall against the full 26-class reference is structurally capped — we cannot match classes that have no entities in the data.

Additionally, 55% of entities remain classified as “Other.” These are primarily measure values (NPS scores, resolution times, coverage amounts) that are correctly preserved as queryable nodes but do not belong to any ontology class. Every comparable ontology induction system (OLLM, AutoSchemaKG, KGGen) excludes literal values from ontology evaluation entirely. Our Entity Assignment metric includes them in the denominator, further depressing the score.

When restricted to the 8 Riskine classes that have evidence in our data (Coverage, Damage, Identification, Property, Risk, Product, Object, Employee), the present-class F1 rises to 0.312, with recall jumping from 0.192 to 0.625 — indicating the pipeline correctly identifies most classes that the data supports.

3.4 Intrinsic Quality (Reference-Free)

Table 4: Intrinsic ontology quality

Metric	Value	Description
Class coverage	32.6%	1,794 of 5,496 entities assigned to a class
Class balance	0.797	Distribution evenness (1.0 = perfectly balanced)
Class coherence	0.430	Avg intra-class cosine similarity
Connectivity	91.6%	Classified entities with cross-class neighbors
Completeness	100%	All classes have inter-class connections
Backbone coverage	100%	All record entities linked to concept entities

3.5 Query Validation

We validated the KG against raw CSV data using analytical queries:

Table 5: KG query results vs. ground truth

Query	KG Result	Ground Truth	Acc.
Longest resolution time	61,606 hours	61,606 hours	100%
Device types by count	Cell Phone: 274	Cell Phone: 287	96%
NPS by claim channel	108 records	116 records	93%
Loss type frequency	3/4 exact	4 types	96%

3.6 Induced Ontology

13 classes induced, with 8 SUBCLASS_OF (IS-A) and 38 ASSOCIATED_WITH (HAS-A) edges:

Table 6: Induced ontology classes

Class	Members	Class	Members
Claim	656	Property	107
Survey	260	Risk	91
Organization	243	Process	47
Person	236	Damage	43
Coverage	166	Activity	27
Document	165	Service	26
Policy	158		

4 Knowledge Graph Summary

Table 7: Final knowledge graph statistics

Metric	Value
Total entities	5,509
Total relationships	27,341
Relationship types	331
Ontology classes	13
SUBCLASS_OF edges	8
ASSOCIATED_WITH edges	38
Decomposed sub-nodes	389
Data sources	9
Duplication rate	0.0%